

Uncovering Patterns in Complex Geometry with Machine Learning

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Abstract

We study dessin d'enfants in this report, combining mathematical analysis with machine learning techniques.³ We begin by examining mathematical foundations of dessin d'enfants, forming the basis for subsequent computational analysis. Using PCA, we try to find the data's dimensionality to a judiciously chosen lower-dimensional space, then apply K-Means clustering to find inherent data patterns. We explore a range of cluster numbers and employ inertia analysis to identify natural clusters. We build a neural network using the MLPClassifier architecture to predict Galois orbit size from dessin representations (adjacency matrices and cyclic edge lists). We achieved an accuracy of 0.42 for adjacency matrices and 0.50 for cyclic edge lists.

Chapter 1

Mathematical aspects

Theorem 1. Let X be a smooth, compact and connected curve over $\mathbb{C}$ with genus g and let $\phi : X \rightarrow \mathbb{P}^1$ be a map of degree d . Then,

$$2g - 2 = -2d + e(\phi) \quad (1.1)$$

where $e(\phi) := \sum_P (e_P - 1)$.⁶

Definition 1. *Belyi map¹* : Let X be a smooth, compact and connected curve over $\mathbb{C}$. A Belyi map (on X) is a nonconstant morphism $\psi : X \rightarrow \mathbb{P}^1$ that is unramified away from $\{0, 1, \infty\}$

An isomorphism of Belyi's maps is a commutative diagram.

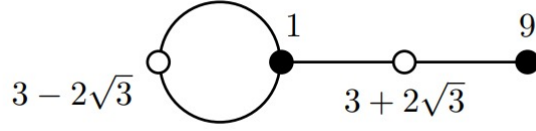
Theorem 2. **Belyi Theorem¹** : A nice curve X over $\mathbb{C}$ admits a Belyi map if and only if X can be defined over a number field $K \subset \mathbb{C}$.

Definition 2. *Dessin⁸* : Given a Belyi map, $\psi : X \rightarrow \mathbb{P}^1$, the preimage $\psi^{-1}([0, 1])$ is a dessin, which is a connected bicolored graph such that two vertices of an edge are colored differently and such that edges around each vertex are given a cyclic ordering. Conversely, a dessin determines a Belyi map, providing an equivalence of categories.

For example, the Belyi map, $\psi : \mathbb{P} \rightarrow \mathbb{P}$ of degree $d = 4$ defined by⁶

$$\psi(x) = -\frac{(x-1)^3(x-9)}{64x} = 1 - \frac{(x^2-6x-3)^2}{64x} \quad (1.2)$$

has preimage 1, 9 for 0 (these are represented by black nodes), and $3 - 2\sqrt{3}, 3 + 2\sqrt{3}$ for 1 (these are represented by white nodes) have the dessin :



ψ being an covering map, is highly surjective, i.e., there could be an arbitrary number of pre-images of $\{0, 1, \infty\}$. We can collect all the ramification indices into a so-called passport, which is the list of ramification indices for each of these pre-images of 0, 1, and ∞ .

Definition 3. Passport : *Passport is the data (g, G, λ) consisting of a non-negative integer $g \in \mathbb{Z}$, a transitive permutation group $G \leq S_d$, and three partitions $\lambda = (\lambda_0, \lambda_1, \lambda_\infty)$ of d .*

The passport of a Belyi map is given by its genus, its monodromy group, and the ramification degrees of the points above $\{0, 1, \infty\}$.

The collection of (x, ψ) gives the Belyi pair which can be used to represent a dessin, i.e., the Belyi maps will be from $X = \mathbb{P}^1$ (with coordinate z) to $\mathbb{P}$ (with coordinate ζ) and will be a rational function in a single complex variable. In this project report, we are only concerned with genus 0, or planar dessins.

In general, Belyi maps are defined over number fields, so the $\text{Gal}(\mathbb{Q}^{al} | \mathbb{Q})$ acts *faithfully* on the set of dessins. But this Galois action on dessins is mysterious and highly unpredictable, and nontrivial invariants are difficult to find.

Some of the dessin representation⁸ for the Belyi map is shown in table 1.1.

$\psi_1(x) = 1 - x^3$		$\psi_2(x) = \frac{(4-x)(1+2x)^2}{27x}$	
$\psi_3(x) = \frac{4(x-1)^3}{27x}$		$\psi_4(x, y) = \frac{x^3+3x^2}{4}$	

Table 1.1: Belyi's map and it's corresponding dessin representation

Definition 4. Galois group⁷ : If E is an extension of the field F and E/F which is an automorphism of E over F is isomorphism. The set of all automorphisms of E/F forms a group with the operation of function composition. This group is called as Galois group.

The combinatorial interpretation of *dessins d'enfants* is related to the absolute Galois group over the algebraic rational numbers. This group describes how these *dessins* transform into each other. The study of dessin d'enfant becomes much more important when one wants to understand the action of absolute Galois group $\Gamma = \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ (the group of automorphisms of the algebraic numbers which fix the rational numbers) of the rational numbers which have no direct description.

Fact 1. There is a Galois action² on algebraic curves defined over $\mathbb{Q}$. Namely, $\sigma \in \Gamma$ acts on a curve by acting on the coefficients of its defining polynomials. This action preserves the genus.

The dessins therefore act as tools to identify invariants and other properties of the rational absolute Galois group. The size of an orbit of a dessin corresponds to the degree of the field extension. Now, it is worth mentioning the theorem about the Galois action.

Theorem 3. The Galois action on dessins² preserves the automorphism group, regularity, regular closures, and the cartographic group.

The orbit of a dessin is all dessin's that the absolute Galois group can transform the original dessin into.

Theorem 4. If A and B are dessins in the same orbit under Γ , then they have the same number of vertices of each colour degree, darts, and faces of each degree. Also, their embedding surfaces have the same genus.

Fact 2. There are finitely many bigraphs for any instance of the parameters of theorem 4

Thus, the orbit of a dessin is finite. We can enumerate all dessins with the same information as in 4 and get a union of finitely many orbits under Γ . The most extensive group of dessins in used dataset represents a quartic extension of the rational numbers. In this context, the quartic extension is associated with a polynomial that possesses four unique roots, affirming its separability. Consequently, each individual dessin within this Galois orbit signifies an extension of the rational numbers, specifically by one of these four distinct roots. These roots play a crucial role in determining the coefficients of the respective Belyi maps associated with these dessins.

Chapter 2

ML techniques on dessin d'enfant

2.1 Dessin Database

We have used the data for 191 dessin d'enfants, available from the work [3]. In this dataset, each dessin is planar, trivalent, and clean (all same colour nodes have the same valency). Out of 191 dessins, 112 dessins have distinct passport in the dataset, but there are 125 orbits of varying degree of extension because there are multiple orbits for some of the passports. When some roots of the minimal polynomials (polynomials which are defined over the field of rational numbers) are complex, then their complex conjugates are also the roots of the polynomials. These pairs of complex roots for polynomials correspond to a dessin, which are isomorphic up to the flipping of their chirality.

When we eliminate redundant matrices arising from isomorphisms, we are left with a set of 152 distinct matrices in the dataset. These matrices represent the 152 unique dessins within their respective orbits, as we consider isomorphisms only within each orbit. Any isomorphisms that exist outside of these orbits are disregarded, and their corresponding matrices are removed. However, in the case of the cyclic edge list representation, all 191 lists remain independent and can be utilized without any reduction.

2.2 Classifying the dessin extension

Now, its the time to do computation on our dessin dataset. But before we start that directly, we need to understand the representation of dessin in Adjacency and Cyclic format.

2.2.1 Adjacency Matrices

To represent a dessin in this format, one basically assigns the numbering to each white nodes (or black nodes). When two nodes are connected by a single edge, we populate both the $\mathcal{M}_{ij}$ and $\mathcal{M}_{ji}$ elements of the matrix with a value of 1. If two edges link the same nodes or if an edge connects a node to itself, these scenarios are

considered as having two connections, resulting 2 being entered in the corresponding $\mathcal{M}_{ij}$ matrix element.

2.2.2 Cyclic Edge lists

To define each dessin in cyclic representation, edges were numbered 1 to 24. White nodes were associated with a clockwise list of three edge numbers, with order being crucial. Black nodes with bivalent nodes, had an anticlockwise list with cyclic ordering being unimportant.

2.3 Machine learning techniques⁵

2.3.1 Principal component analysis (PCA)

Principal Component Analysis (PCA)⁴ is a statistical technique introduced by Karl Pearson in 1901. It aims to simplify data by reducing its dimensionality while preserving important information. PCA transforms a set of related variables into a new set of uncorrelated variables through a mathematical process. PCA is an unsupervised learning method used to explore relationships among variables. It helps find a more compact representation of the data without needing prior knowledge of target variables.

Principal Component Analysis (PCA) is a technique that finds a set of orthogonal axes, known as principal components, that capture the most significant variations in the data. These components are combinations of the original variables and are arranged in decreasing importance. The total variance explained by these components equals the variance in the original data.

For our dataset of *dessin*, we basically, flattened the matrices (adjacency matrices), and then computed the covariance matrix, calculated eigen values, sorted it, and then projected the data onto 2D, to see the match.

Table 2.1: Table showing projection of data for adjacency matrix. N_λ is number of eigenvectors to which data is projected and $\bar{\sigma}$ is explained variance ratio corresponding to N_λ .

N_λ	$\bar{\sigma}$
2	0.26012
3	0.34548
20	0.96705
25	0.99400
30	0.99952

Clearly, from the table 2.1, 2D projection just constitutes about 26 %. Moreover, as the eigenvectors increases for projecting our data, then accuracy also increases, which is quite obvious. The graph showing the simple 2D projection as well as with the corresponding ‘Galois orbit size’ is shown in figure 2.2 and 2.3, respectively. We apply PCA to our *dessin d’enfant* dataset, to find what dimension suits our data.

This can also be done for cyclic dataset, where we got more accuracy than this. Table 2.2 clearly shows

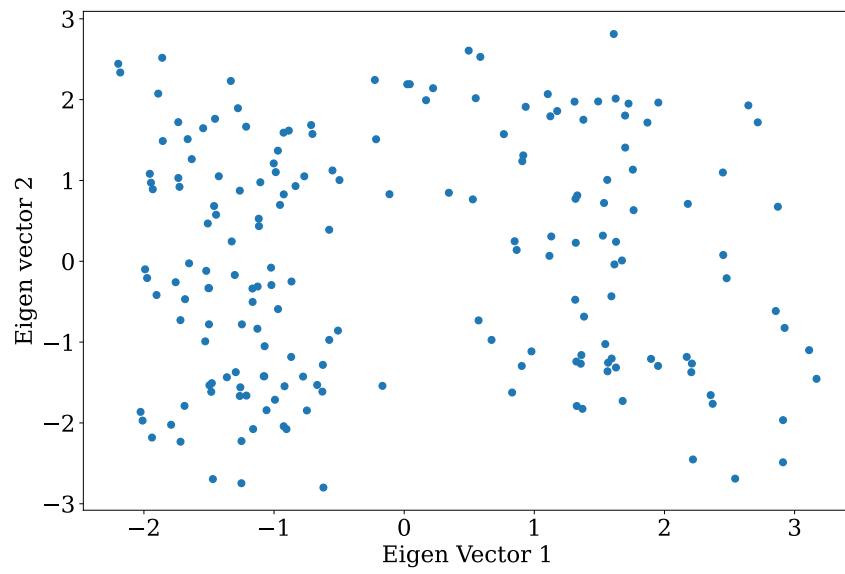


Figure 2.1: Plot showing 2D projection of adjacency matrix

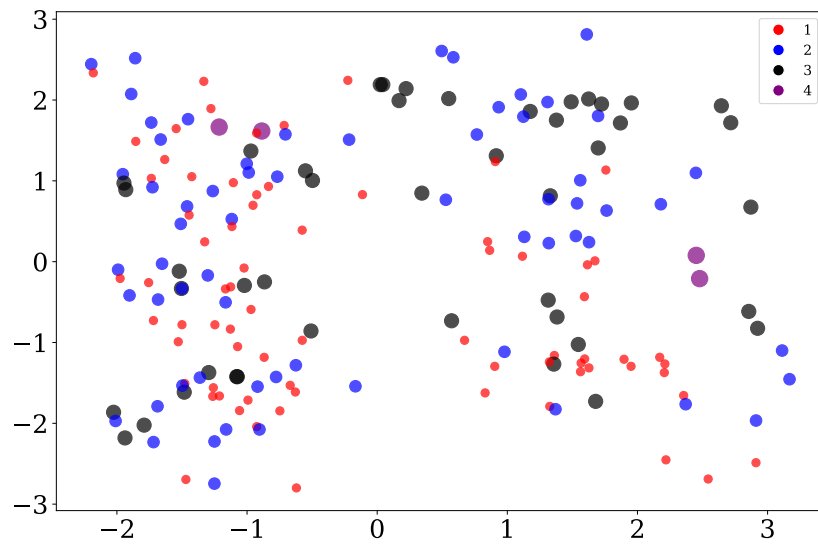


Figure 2.2: Plot showing 2D projection of adjacency matrix according to Galois orbit size

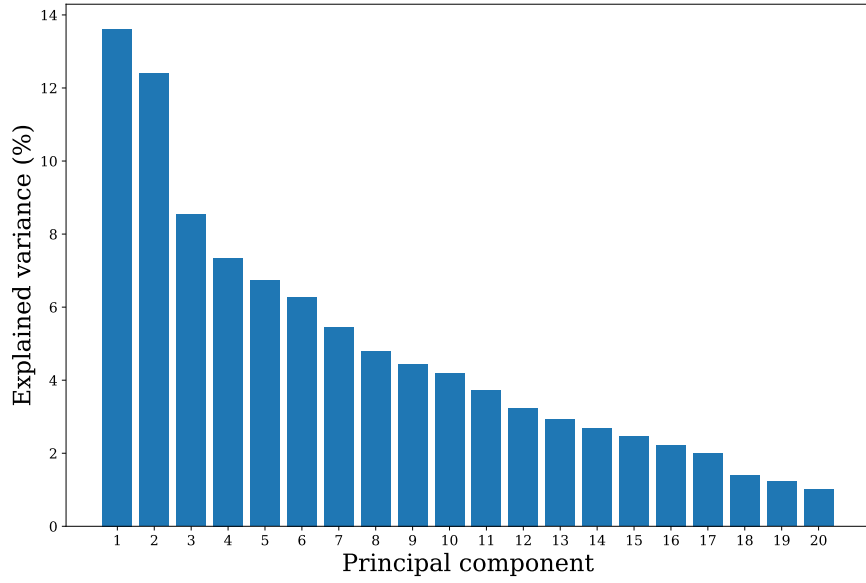


Figure 2.3: Histogram showing the % explained variance vs number of principal component

Table 2.2: Table showing projection of data for cyclic representation of dessin

N_λ	$\bar{\sigma}$
2	0.35733
3	0.46382
5	0.63519
10	0.87353
20	0.99576

that projection over cyclic eigen vectors accounts for better accuracy than adjacency matrix representation for the dessin for the same number of eigenvectors. Also, through these plot we tried to predict whether PCA can separate the dessins according to orbit size, or not. Clearly, from figure 2.2 and 2.7 telling us that the relation between the dessins and orbit size is not linear.

2.3.2 K-Means clustering

K-means clustering is a widely used unsupervised machine learning algorithm[5] for partitioning a dataset into distinct groups or clusters based on the similarity of data points. It is a valuable technique in data analysis and pattern recognition. K-means aims to find natural groupings within data, making it a highly useful tool in various applications including segmentation, image compression, etc.

In K-means clustering, the “elbow method” is often used to determine the optimal number of clusters (k) for a given dataset. It involves plotting the within-cluster sum of squares (WCSS, It’s the measure of total variance within the clusters, also called as Inertia.) as a function of the number of clusters and looking for the “elbow point” where the rate of decrease in WCSS starts to slow down. This point can indicate the appropriate number

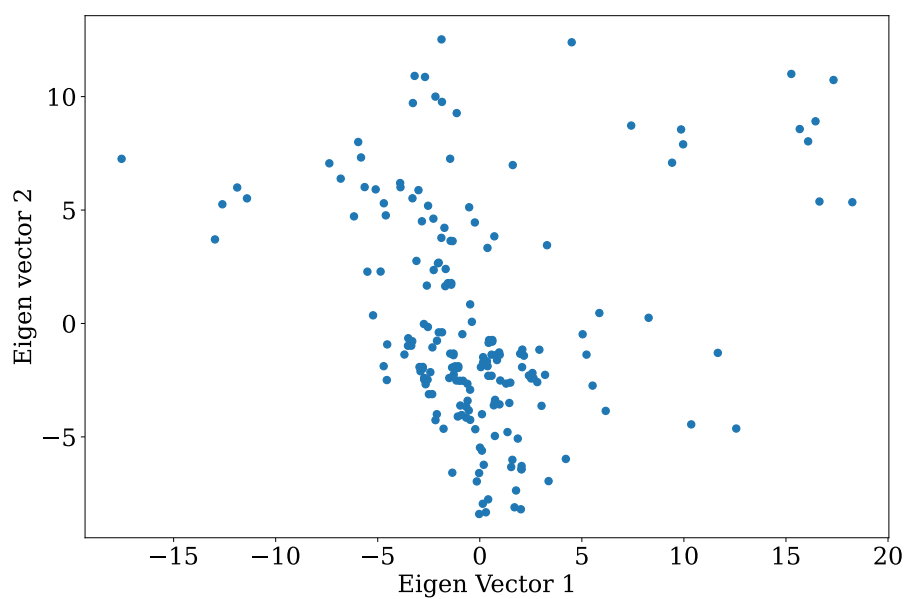


Figure 2.4: Plot showing 2D projection for cyclic representation of dessin

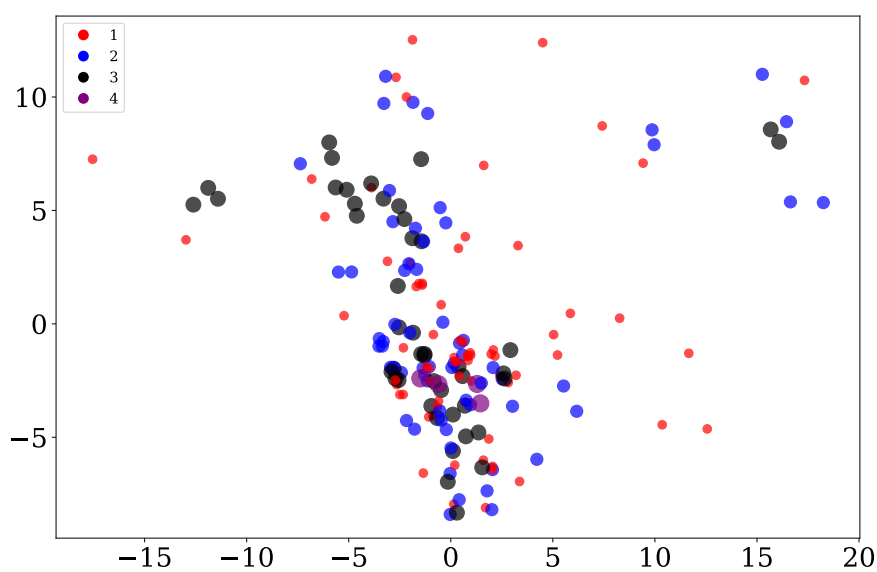


Figure 2.5: Plot showing 2D projection for cyclic representation of dessin according to there Galois orbit size.

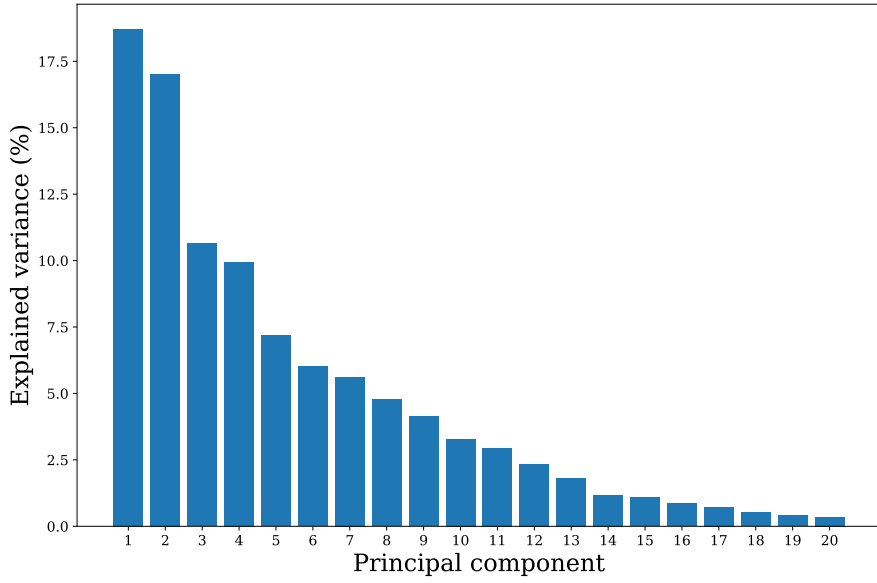


Figure 2.6: Histogram showing the % explained variance vs number of principal components for cyclic representation of the dessin.

of clusters for your data.

We applied K-Means clustering to our *dessin d'enfant* dataset for adjacency matrix, and the figure 2.7 shows the elbow point ($k=3$) for the same. We then applied the K-Means clustering on cyclic dataset, whose graph is also plotted and shown in figure 2.8. Here, the "elbow point" is not as distinct as in the case of 2.6. It appears to be around $k=3$ or $k=4$, but it's somewhat subjective.

2.4 Neural Networks

In this work, we employ a powerful neural network⁵ to address the task of learning the Galois orbit size from flattened dessin representations. The neural network architecture comprises input, hidden, and output layers, where the input layer receives flattened dessin data, and the hidden layers extract features and apply activation functions. Hyperparameter tuning, involving adjustments to the number of hidden layers, nodes per layer, learning rate, and activation functions, plays a pivotal role in optimizing the network's performance. This approach offers a unique bridge between mathematical theory and data-driven analysis, with potential applications in algebraic and geometric research, where precise measurements of Galois orbit size are of paramount importance. By fine-tuning hyperparameters and optimizing architecture, we aim to obtain high accuracy.

We employed a neural network to predict Galois orbit sizes based on flattened dessin representations. The first step involves data preparation, where we segregate the dataset into dessin representations and corresponding

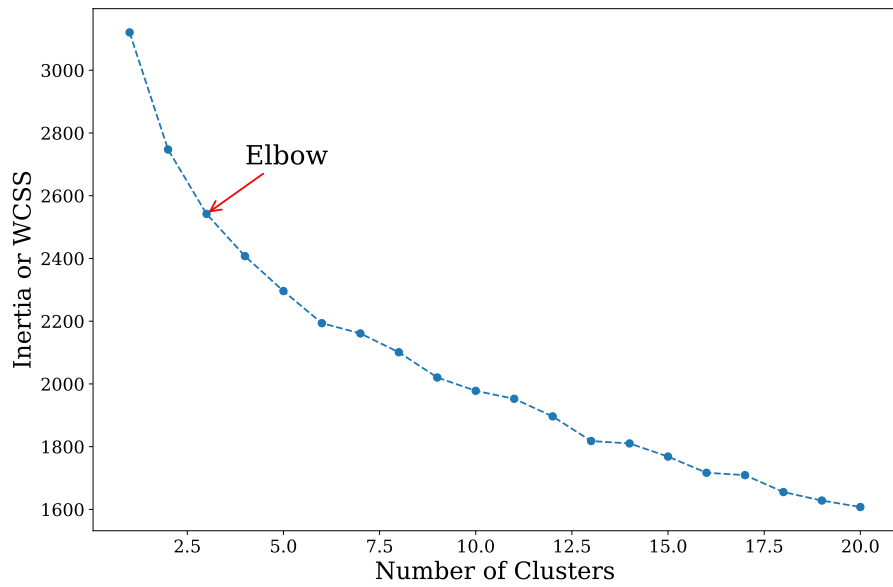


Figure 2.7: Plot showing the inertial values against number of clusters for adjacency matrix representation

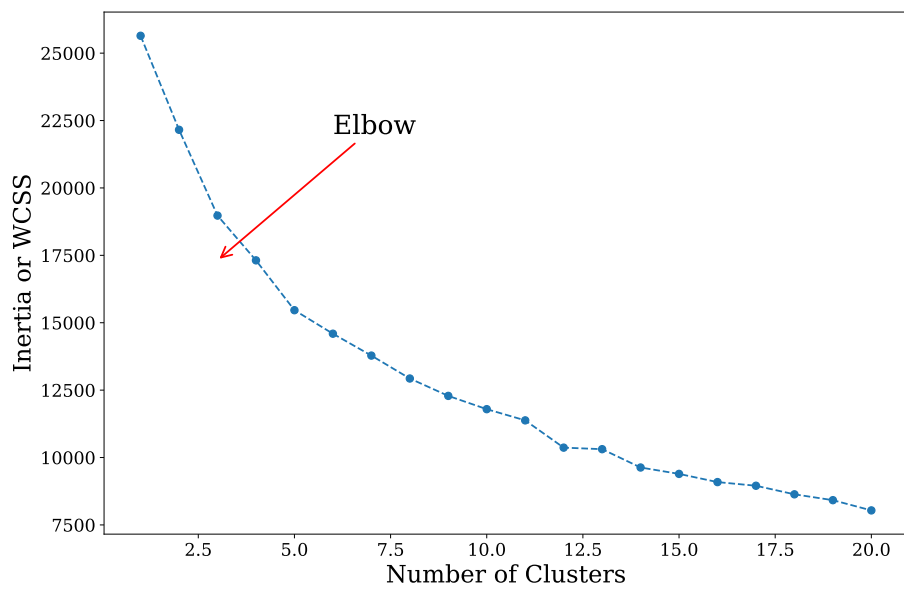


Figure 2.8: Plot showing the inertial values against number of clusters for cyclic representation of dessin.

Galois orbit sizes. Subsequently, we partition the data into three distinct sets: a training set, a validation set, and a test set. This partitioning facilitates model development and evaluation. The neural network chosen for this task is an MLP (Multi-Layer Perceptron) classifier. Notably, we incorporate a rigorous hyperparameter tuning process using grid search with cross-validation ($cv = 3$). This systematic exploration encompasses variations in essential factors, including the number of hidden layers, nodes per layer, and learning rates. We then proceed to train the MLP classifier using the training data. The MLP is configured with a single hidden layer of 100 nodes and uses the ReLU (Rectified Linear Unit) activation function. The 'adam' solver is used for optimization, and the model is allowed a maximum of 1000 iterations for learning. Following training, the model's performance is thoroughly assessed using the test data, with accuracy as our primary metric. The model achieves an accuracy of approximately 42.1 % (for adjacency matrix) and 50 % (for cyclic representation), indicating its capacity to discern intricate patterns within the flattened dessin representations and their relationship to Galois orbit sizes. However, it's worth noting that further refinements or alternative strategies may hold potential for enhancing predictive accuracy.

Table 2.3: Table showing accuracy and standard deviation (σ) of the NN model

	Accuracy	σ
Adjacency	0.42	0.2189
Cyclic	0.50	0.1823

Bibliography

- [1] G. V. Belyi. “On Galois Extensions of a Maximal Cyclotomic Field”. In: *Mathematics of the USSR-Izvestiya* 14.2 (1980), p. 247.
- [2] Carson Collins. *The Galois Action on Dessin d’Enfants*. Tech. rep. University of Chicago, 2019. URL: <http://math.uchicago.edu/~may/REU2019/REUPapers/Collins.pdf>.
- [3] Yang-Hui He, Edward Hirst, and Toby Peterken. “Machine-Learning Dessins d’Enfants: Explorations via Modular and Seiberg-Witten Curves”. In: *J. Phys. A: Math. Theor.* ().
- [4] Tom Howley et al. “The Effect of Principal Component Analysis on Machine Learning Accuracy with High Dimensional Spectral Data”. In: Jan. 2005.
- [5] F. Pedregosa et al. “Scikit-learn: Machine Learning in Python”. In: *Journal of Machine Learning Research* 12 (2011), pp. 2825–2830.
- [6] *VaNTAGe: A Virtual Math Seminar on Open Conjectures in Number Theory and Arithmetic Geometry*. <https://sites.google.com/view/vantageseminar/history-of-vantage-talks>. The 9th topic is Belyi maps and Hurwitz spaces. 2021.
- [7] Wikipedia contributors. *Galois Group*. URL: https://en.wikipedia.org/wiki/Galois_group.
- [8] Leonardo Zapponi. “What is Dessin d’Enfants”. In: *Notices of the American Mathematical Society* 50.7 (2003), pp. 705–707. URL: <https://www.ams.org/notices/200307/what-is.pdf>.